

**PARTS LIST***Semiconductors:*

IC1 (IC1A, IC1B, IC1E)	- LM324 quad op-amp
IC2 (IC2A-IC2E)	- 74C00 quad NAND gate
T1	- BC547 NPN transistor
D1-D7	- 1N4007 rectifier diode
LED1	- 5mm LED red colour

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 4.7k Ω
R2, R7	- 3.3k Ω
R3, R4, R6	- 10k Ω
R5	- 5.6k Ω
R8	- 2.2k Ω
R9	- 1k Ω
VR1, VR2	- 5k Ω potentiometer

Capacitors:

C1-C3	- 0.01 μ F ceramic disk
C4	- 0.1 μ F ceramic disk

Miscellaneous:

S1	- Push-to-on switch
CON1	- 2-pin connectors
PZ1	- 12V piezo buzzer
	- Twin-core cable (required length)

is built around quad op-amp LM324 (IC1), quad 2-input NAND gate IC 7400 (IC2), and a few other components.

The quad op-amp IC LM324 (IC1) functions as a voltage comparator. The IC1A comparator detects higher voltage output from the sensor (in the dotted box) when the connecting wires (between TP1 and TP2) are short-circuited or connected while

someone attempts to tamper with the vehicle.

The IC1B comparator detects lower voltage when the sensor wires are cut or become open-circuited. Potentiometers VR1 and VR2 provide reference voltages for comparators IC1A and IC1B. Under normal conditions, both comparators produce logic low outputs. However, during voltage disturbances, the output of either comparator becomes logic high and initiate the alarm.

Construction and testing

An actual-size, single-sided PCB layout for the circuit is shown in Fig. 3, and its component layout in Fig. 4. Alternatively, the circuit can be assembled as per Fig. 2 on a multipurpose board.

Once assembled, the mounted board should be enclosed in a suitable enclosure with openings for the buzzer and LED indicator. A PVC electrical box, readily available at low cost, is a good option. The sensor circuit can be soldered onto a strip of multipurpose board with sturdy leads to connect the main circuit wires.

For testing, connect the sensor (within dotted lines in the circuit diagram) to the main circuit PCB. Power the circuit using a 12V DC adaptor and a 9V backup battery.

During calibration

- Adjust potentiometer VR1 to 3.85V and VR2 to 2.25V
 - Connect the security wire loop between TP1 and TP2
 - Open, short and reverse; ensure buzzer PZ1 and LED1 turn on
 - Press push button S1 momentarily to turn off the buzzer and LED after connecting the security loop between TP1 and TP2

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